

Installation Guide

- Linux and MacOS: <https://www.cse.iitb.ac.in/~mythili/os/xv6.html>
- Make sure you are able to compile and run xv6 at this point. You can write a simple user program and run it to check if everything works. Follow the section “Writing and testing new code” in the given link for this.

Reference Book

<https://pdos.csail.mit.edu/6.828/2025/xv6/book-riscv-rev5.pdf>

You may use this book for conceptual understanding on topics relevant to your project. You can also refer to the section titled “PART X: The xv6 operating system” in [this course webpage](#).

Tutorial 01: How to implement a new system call into xv6?

- [Operating Systems Lecture 23: System calls for process management in xv6](#)
- [xv6 -- Adding a system call](#)

In this project, you will form a group of two students. Your task is to implement **at least two additional features** into the xv6 operating system. At the beginning, you will submit a 2-page project proposal describing the features you want to implement. Upon approval from the instructor, you will work on the features and submit a report at the end of the semester. You will also be asked to demonstrate your implementation with a short presentation.

A few project ideas you can refer to:

- <https://github.com/remzi-arpacidusseau/ostep-projects>.
- <https://pages.cs.wisc.edu/~remzi/OSTEP/lab-projects-xv6.pdf>

Please note that this is a self-driven project. You are expected to learn and solve problems both individually and as a group. You can consult with the instructor or the TA if you have any issues. However, while we may provide some general guidance, you have to explore and work through most challenges on your own.

Project timeline:

Deadlines	Tasks
	• Formation of groups • Get familiar with xv6 using the shared resources
08.03.2026	• Submit project proposal (max 2 pages) and get feedback • Show that you have implemented a system call successfully
	• Update-1: At least one task should be implemented
	• Update-2: Both tasks should be implemented
	• Submit final report via email (max 6 pages)